Face Authentication Under Presentation and Adversarial Attacks

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Course : Biometrics (CS414)

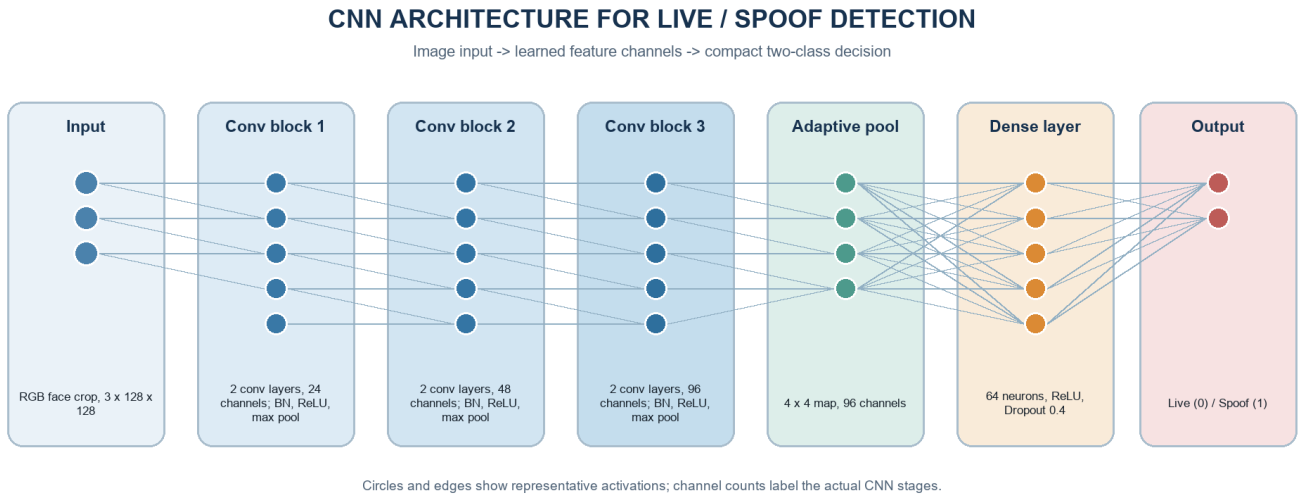


Figure 1. CNN pipeline with representative neuron activations and connections.

1. Abstract

Face presentation attack detection (PAD) helps an authentication system reject a photograph or replay presented to a camera as if it were a live person. This project trains a compact convolutional neural network to classify face images as Live or Spoof. The model learned from 1,361 OULU training images, selected its checkpoint at epoch nineteen using a subject-held-out OULU validation set, and was evaluated on clean and FGSM-perturbed images from CASIA-FASD and a balanced CelebA-Spoof sample. OULU validation reached 98.8% accuracy and 98.7% balanced accuracy. On clean external tests, balanced accuracy was 49.7% on CASIA-FASD and 54.5% on CelebA-Spoof. With FGSM perturbations at epsilon 8/255, balanced accuracy fell to 16.6% and 5.7%, respectively. The training and evaluation pipeline completed, but the detector did not generalize reliably across datasets and was not robust to the tested perturbation. Replay-Mobile could not be scored because the local videos had no labels.

2. Description and Introduction

A presentation attack attempts to fool face authentication by showing a printed photograph, a replayed video, or another artificial face to the camera. A PAD model checks whether an input appears live before access is granted; it does not identify the person. Assignment 39 asks for training on OULU-NPU, validation on Replay-Mobile, testing on CASIA-FASD and CelebA-Spoof, and adversarial evaluation. The available local Replay-Mobile files are unlabeled, so the experiment uses a subject-separated OULU holdout for scored validation and documents this deviation.

3. Datasets Used

Only OULU images were used to fit the model. CASIA-FASD and CelebA-Spoof were kept as external test sources. These datasets provide training examples, a labeled validation set where available, and independent sources for external testing.

Dataset	Available labeled data	Use in this experiment
OULU-NPU local subset	1,701 stills: 343 Live, 1,358 Spoof; 20 subjects	Training and subject-held-out validation
Replay-Mobile local sample	39 MP4 videos; no Live/Spoof labels found	Not scored; cannot compute validation metrics
CASIA-FASD test	65,786 images: 10,128 Live, 55,658 Spoof	Full local test folders; clean and FGSM evaluation
CelebA-Spoof intra-test	67,170 label rows: 19,923 Live, 47,247 Spoof	Balanced sample of 1,000 per class

3.1 OULU split and labels

The local OULU folders use true for Live (label 0) and false for Spoof (label 1). The third underscore-separated filename field supplies the subject ID. A fixed seed selected four of 20 subjects for validation and kept each subject entirely on one side of the split. Training used 1,361 images from 16 subjects (276 Live and 1,085 Spoof); validation used 340 images from four subjects (67 Live and 273 Spoof). The downloaded local subset has no official protocol split files, so these counts describe the available copy rather than the full OULU-NPU protocol.

3.2 Replay-Mobile limitation

The local Replay-Mobile directory contains 39 videos grouped by device pair, but no class annotations or split metadata identifying live and spoof clips. The videos cannot be scored without guessing labels. The OULU subject holdout is therefore used as the validation set, and results are not presented as Replay-Mobile performance.

4. Methodology and Implementation

The notebook reads images as RGB, resizes them to 128 x 128 pixels, and scales pixel values to the 0-1 range. Images are processed in batches so the full image collections are not loaded into memory. Training augmentation uses random horizontal flips and small brightness/contrast changes. A square-root inverse-frequency sampler partly balances the Live/Spoof training distribution.

4.1 CNN architecture

Stage	Configuration
Input	RGB image, 3 x 128 x 128
Convolution block 1	Two 3 x 3 convolutions, 24 channels; batch normalization, ReLU, max pooling
Convolution block 2	Two 3 x 3 convolutions, 48 channels; batch normalization, ReLU, max pooling
Convolution block 3	Two 3 x 3 convolutions, 96 channels; batch normalization, ReLU, max pooling
Feature reduction	Adaptive average pooling to 4 x 4; flatten
Classifier	Dense 64, ReLU, dropout 0.4, then two Live/Spoof logits

4.2 Training setup

Setting	Configuration
Training source	1,361 OULU images from 16 subjects
Input size and batch	128 x 128 RGB; batch size 16
Optimization	Cross-entropy loss; AdamW; learning rate 0.0005; weight decay 0.0001
Run	20 epochs; fixed seed 42; CUDA if available, otherwise CPU
Augmentation and sampling	Horizontal flip; brightness/contrast 0.9-1.1; square-root weighted sampler
Selected checkpoint	Epoch 19, highest OULU validation balanced accuracy: 98.7%

4.3 Evaluation protocol

This report focuses on accuracy, balanced accuracy, and recall for each class. Accuracy is the share of all images classified correctly. Balanced accuracy is the average of Live recall and Spoof recall, so each class contributes equally even when their counts differ. Live recall measures how often genuine images are correctly accepted as Live; Spoof recall measures how often attack images are correctly rejected as Spoof. The confusion matrices provide a count-based view: rows show the true class and columns show the model prediction. For adversarial evaluation, the Fast Gradient Sign Method (FGSM) adds a small, input-dependent perturbation with epsilon 8/255, then clips pixel values to the valid image range. The model was not trained on adversarial examples.

5. Results with Graphs and Analysis

Table 1 compares the model's clean-image validation result with its performance on two external datasets. The FGSM rows show how the same external test images were classified after a small adversarial perturbation. Accuracy gives an overall summary, while balanced accuracy is more informative when class counts are uneven.

Dataset	Condition	Images	Accuracy	Balanced accuracy
OULU validation	Clean	340	98.8%	98.7%
CASIA-FASD test	Clean	65,786	72.5%	49.7%
CASIA-FASD test	FGSM	65,786	28.0%	16.6%
CelebA-Spoof	Clean	2,000	54.5%	54.5%
CelebA-Spoof	FGSM	2,000	5.7%	5.7%

Table 1. Overall classification performance on clean and adversarial images.

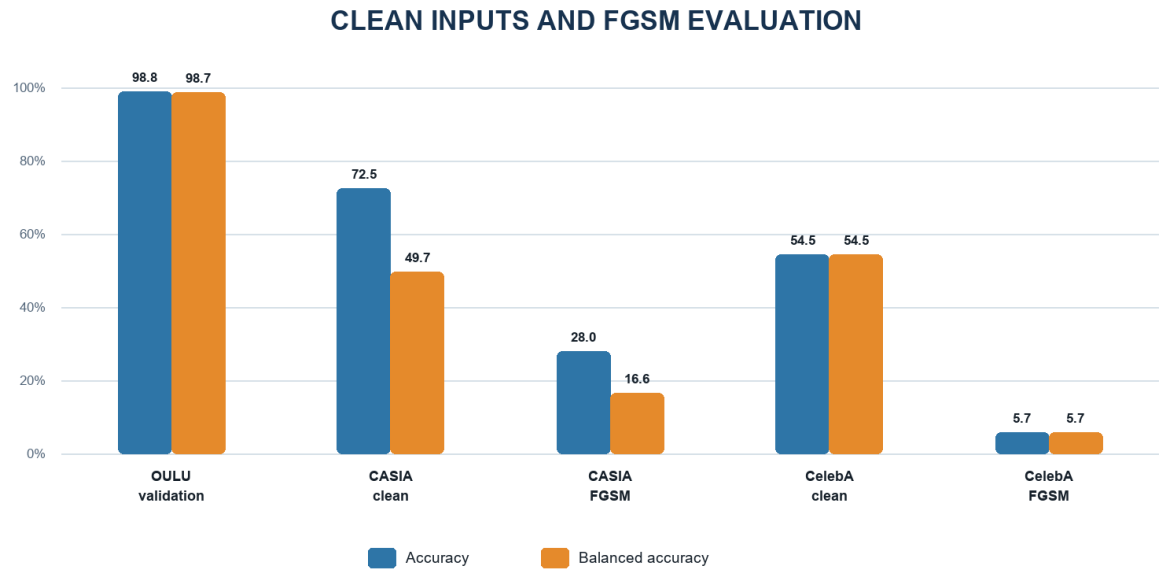


Figure 2. Accuracy and balanced accuracy across the evaluation sets.



Figure 3. Training loss and OULU validation balanced accuracy over twenty epochs.

5.1 Understanding the class measures

Accuracy is the percentage of all images that received the correct label. It can be misleading when one class is much more common than the other.

Balanced accuracy addresses this by averaging Live recall and Spoof recall.

Live recall is the proportion of genuine images classified as Live, while Spoof recall is the proportion of attack images classified as Spoof.

For a face authentication system, both matter: low Live recall rejects genuine users, and low Spoof recall allows attacks through.

Dataset	Live recall	Spoof recall
OULU validation	98.5%	98.9%
CASIA-FASD test	16.7%	82.7%
CelebA-Spoof	50.7%	58.2%

Table 2. Recall for each class, showing which type of image the detector recognizes.

5.2 Interpretation of the results

The OULU validation result is high, but it is an optimistic checkpoint-selection estimate: the same 340-image holdout was used to choose the best of 20 epochs. It is not an independent final test. The training loss fell overall from 0.4837 to 0.0736, while validation balanced accuracy fluctuated across epochs. The validation set is small, with 67 Live images, so a few changed predictions can shift the score.

External performance is weak. CASIA-FASD contains 84.6% Spoof images, so predicting every image as Spoof would yield about 84.6% ordinary accuracy. The model reached 72.5% accuracy and 49.7% balanced accuracy. Its Live recall was only 16.7%, meaning it missed most genuine live images. CelebA-Spoof was sampled evenly, and its 54.5% accuracy/balanced accuracy is only slightly better than chance.

But, FGSM further degraded performance: CASIA balanced accuracy fell to 16.6%, while CelebA-Spoof fell to 5.7%. The model is therefore not robust to this attack. The high OULU score does not transfer to the different image sources and capture conditions in the external datasets.

5.3 Confusion matrices

The matrices below summarize the classifications for each dataset and condition. Each cell shows the number of images and the percentage within its actual-class row.

Spoof images placed in the predicted Live column are missed attacks;

Live images placed in the predicted Spoof column are genuine users incorrectly rejected.



Figure 4. Confusion matrices.

6. Conclusion

The CNN training, checkpoint selection, clean evaluation, and FGSM evaluation completed on the available labeled images. The system scored highly on the OULU validation holdout but transferred poorly to CASIA-FASD and CelebA-Spoof, and its scores collapsed under FGSM. The notebook is a working baseline and demonstrates the complete evaluation pipeline; the trained detector is not reliable enough for face authentication.

The local OULU subset is small, so the experiment only partially matches the assignment's intended validation protocol.

7. Future Work

- Obtain the official Replay-Mobile labels and evaluate the required validation split.
- Use a development set for checkpoint selection and reserve an untouched test set for the final estimate.
- Add more labeled OULU subjects and attack conditions; test a lightweight pretrained backbone.
- Evaluate video-level aggregation and report PAD metrics.
- Try adversarial training or another robustness method, then evaluate on held-out FGSM examples.